```
# 1. Import required libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from sklearn.datasets import load_iris
```

```
# 2. Load the Iris dataset
iris = load_iris(as_frame=True)
df = iris.frame
```

```
# Add flower species names
df["species"] = df["target"].map(
    dict(enumerate(iris.target_names))
)
```

```
# 3. Check for missing values
print("Missing Values:")
print(df.isnull().sum())
```

```
Missing Values:
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
target               0
species              0
dtype: int64
```

```
# 4. Display column names
print("\nColumn Names:")
print(df.columns.tolist())
```

```

Column Names:
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)', 'target', 'species']
```

```
# 5. Display dataset shape
print("\nDataset Shape:", df.shape)
```

```

Dataset Shape: (150, 6)
```

```
# 6. Remove rows containing missing values
df.dropna(inplace=True)
```

```
# 7. Remove unimportant feature (target is redundant
# because species contains the class names)
df.drop(columns=["target"], inplace=True)
```

```
print("\nCleaned Dataset Shape:", df.shape)
print("\nDataset Information:")
print(df.head())
```

```

Cleaned Dataset Shape: (150, 5)

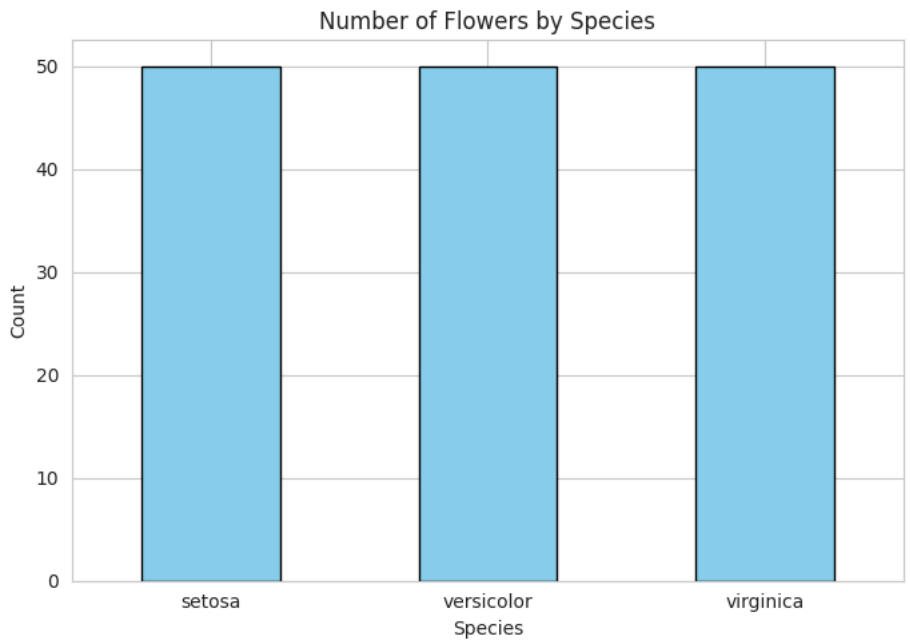
Dataset Information:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

  species
0  setosa
1  setosa
2  setosa
3  setosa
4  setosa
```

```
# Set Seaborn style
sns.set_style("whitegrid")
```

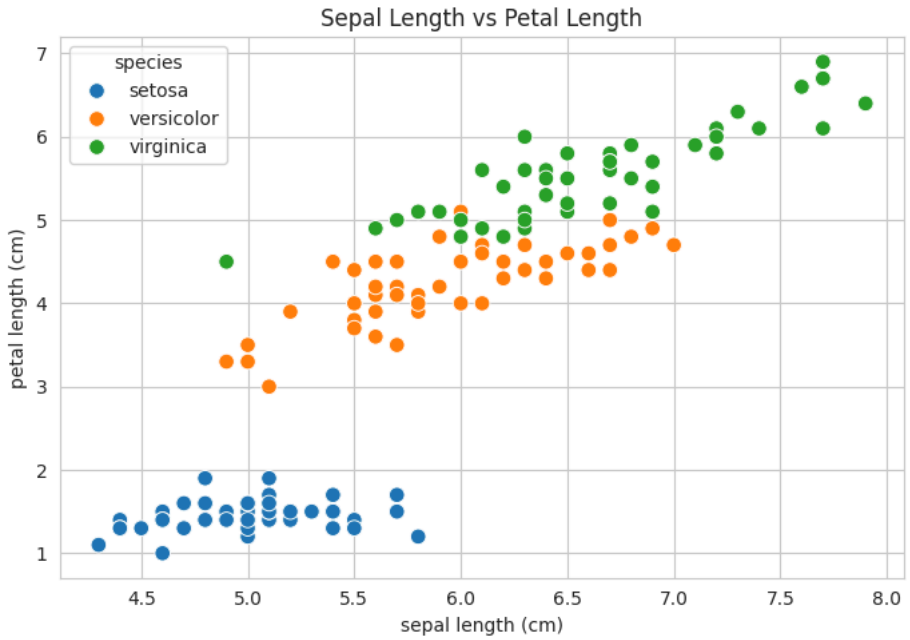
BAR PLOT

```
# 8(a). BAR PLOT using Matplotlib
plt.figure(figsize=(7, 5))
df["species"].value_counts().plot(
    kind="bar", color="skyblue", edgecolor="black"
)
plt.title("Number of Flowers by Species")
plt.xlabel("Species")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



SCATTER PLOT

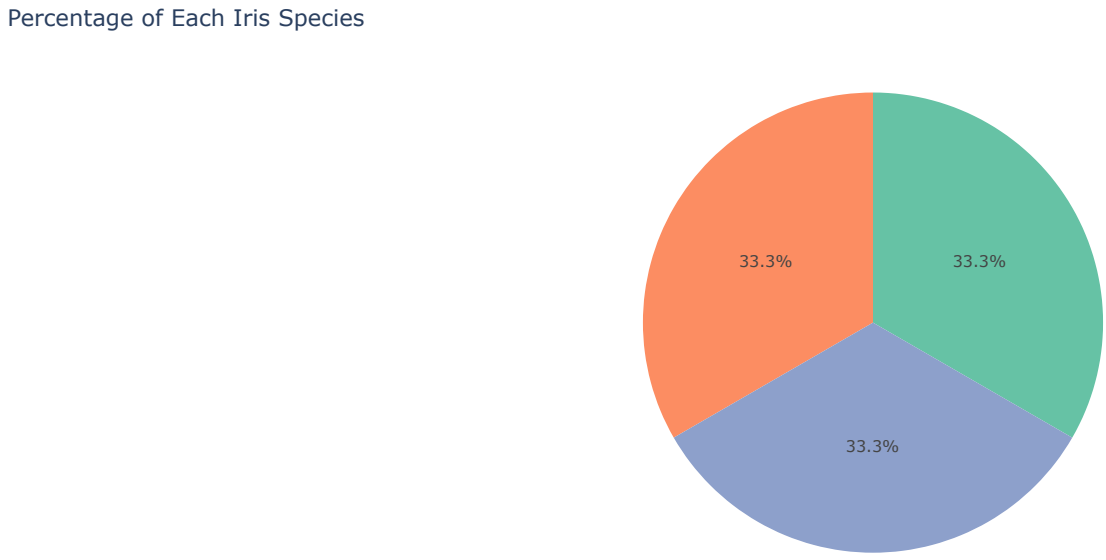
```
# 8(b). SCATTER PLOT using Seaborn
plt.figure(figsize=(7, 5))
sns.scatterplot(
    data=df,
    x="sepal length (cm)",
    y="petal length (cm)",
    hue="species",
    s=70
)
plt.title("Sepal Length vs Petal Length")
plt.tight_layout()
plt.show()
```



PIE CHART

```
# 8(c). PIE CHART using Plotly
species_counts = df["species"].value_counts().reset_index()
species_counts.columns = ["species", "count"]

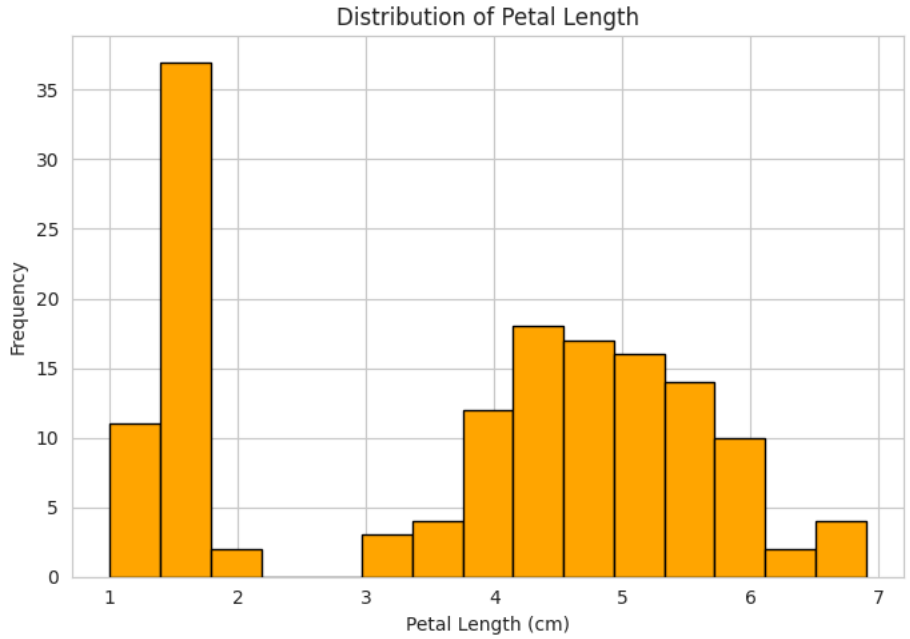
fig = px.pie(
    species_counts,
    names="species",
    values="count",
    title="Percentage of Each Iris Species",
    color_discrete_sequence=px.colors.qualitative.Set2
)
fig.show()
```



HISTOGRAM bold text

```
# 8(d). HISTOGRAM using Matplotlib
plt.figure(figsize=(7, 5))
plt.hist(
    df["petal length (cm)"],
```

```
    bins=15,
    color="orange",
    edgecolor="black"
)
plt.title("Distribution of Petal Length")
plt.xlabel("Petal Length (cm)")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



BOX PLOT

```
# 8(e). BOX PLOT using Seaborn
plt.figure(figsize=(7, 5))
sns.boxplot(
    data=df,
    x="species",
    y="petal length (cm)"
)
plt.title("Petal Length Distribution by Species")
plt.xlabel("Species")
plt.ylabel("Petal Length (cm)")
plt.tight_layout()
plt.show()
```

